

Khawaja Yunus Ali University

School of Science and Engineering
Department of Computer Science and Engineering

Mid-term Examination: Summer 2022

Program: B.Sc. in CSE (Batch – 14th) 1st Year 1st Semester

Course Title: Physics - I

Course Code: PHY 1101

Time: 10 Minutes

Total Marks: 10 X 1 = 10

Student ID:	Batch:	Session:
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Part 1: MCQ

These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.

- The net flux of a cylindrical Gaussian surface placed in an external field is
 - positive.
 - negative.
 - infinity.
 - one.
 - zero.
- What is the linear charge density of the wire of length 200cm having 1000C charge?
 - 500C/m
 - 50C/m
 - 5C/m
 - 2C/m
 - 0.5C/m
- The flux will be maximum at angle between $\vec{E}$ and $\vec{da}$ is
 - -60° .
 - -30° .
 - 0° .
 - 45° .
 - 90° .
- What is the unit of resistivity?
 - Farad
 - Volt
 - Ohm
 - Ohm-meter
 - Joule
- If the cross-sectional area is double, then the new drift velocity will
 - become 4 times.
 - become 2 times.
 - become half.
 - become zero.
 - remain same .

P.T.O.

6. 1 Farad = ?
- a) $1\text{C}/1\text{V}$
 - b) $1\text{V}/1\text{C}$
 - c) $1\text{C}/1\text{Sec}$
 - d) $1\text{J}/1\text{C}$
 - e) $1\text{N}/1\text{C}$
7. The value of the permittivity of the free space is
- a) $1.6 \times 10^{-16} \text{Nm}^{-1}$.
 - b) $8.85 \times 10^{-12} \text{C}^2 \text{N}^{-1} \text{m}^{-2}$.
 - c) $9 \times 10^9 \text{Nm}^2 \text{C}^{-2}$.
 - d) $6.6 \times 10^{-16} \text{Nm}^{-1}$.
 - e) $1.6 \times 10^{-19} \text{Nm}^{-1}$.
8. The force that a charge experiences while moving simultaneously through the electric field and magnetic field is called force.
- a) electric
 - b) Coulombs
 - c) Newtonian
 - d) Lorentz
 - e) magnetic
9. The potential differences between two points in an equipotential surface is
- a) 4.
 - b) 3.
 - c) 2.
 - d) 1.
 - e) 0.
10. The inward direction of the magnetic field in the plane of paper indicated by sign.
- a) dot
 - b) cross
 - c) plus
 - d) minus
 - e) arrow

Invigilator Signature

Examiner Signature

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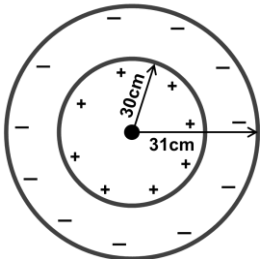
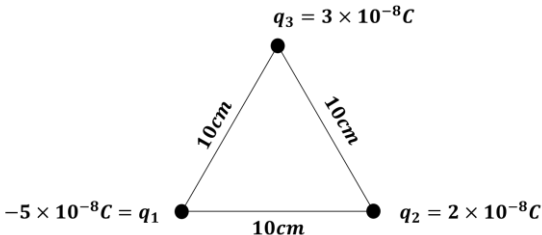
Time: 1 Hour and 20 Minutes

Total Marks: 20

Part 2: SAQ

(Answer Question No. 1 and any three from the rest.)

[The figures in the right margin indicate full marks for the respective question]

1.	See the stem carefully and answer the question that follows: A capacitor consisting of two concentric spheres –		
	 Potential difference between two spheres is 500V		
	a) How much energy will be stored in the capacitor?	[2]	CLO2
	b) Compare the amount of free charge and induced charge, if dielectric material ($K=1.5$) is placed between two plates.	[3]	CLO4
2.	a) Define electric flux by surface integration.	[1]	CLO1
	b) Formulate the Coulomb's law from Gauss law.	[2]	CLO2
	c) A silver wire 1 mm in diameter carries a charge of 90 C in 1 hr. Silver contains 5.8×10^{28} free electrons/m ³ . Calculate – (i) the current density of the wire. (ii) the drift velocity of electrons in the wire.	[2]	CLO4
3.	a) Identify the limitations of Coulomb's law.	[1]	CLO1
	b) Relate electric field with charge density and current density.	[2]	CLO1 & CLO4
	c) Calculate the electric field strength at a point when there is an electron experienced $89.18 \times 10^{-31} N$ force.	[2]	CLO2
4.	See the stem and answer the question that follows:		
			
	a) Illustrate if there is no q_1 charge in the above system then what will be the structure of electric lines of force?	[2]	CLO3 & CLO4
	b) Find the electric potential energy for the above system.	[3]	CLO1 & CLO2
5.	a) State Fleming's right-hand rule.	[1]	CLO1
	b) Summarize the Oersted's Experiment.	[2]	CLO4
	c) A straight horizontal segment of copper wire carries $I = 28A$. What is the magnitude of the magnetic field needed of float the wire, that is, to balance its weight? Its linear mass density is 46.6g/m.	[2]	CLO2